



Cauhoiktpm - câu hỏi ôn tập kiểm thử phần mềm cuối kỳ

VKU_Đảm bảo chất lượng và kiểm thử phần mềm (Trường Đại học Công nghệ Thông tin và Truyền thông Việt - Hàn)



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How is measurement used in software quality assurance?

- A. **To track software quality and assess the effectiveness of methodological and procedural changes**
- B. To reduce the number of developers needed
- C. To replace testing with automated tools
- D. To increase the speed of code writing

What is the main objective of conducting formal technical reviews?

- A. To increase team bonding
- B. To speed up the development process
- C. **To uncover quality problems**
- D. To train new developers

What is the primary goal of review reporting and record keeping?

- A) To keep track of the budget
- B) **To document the review process and ensure that issues are identified and addressed**
- C) To promote the product to potential customers
- D) To create a training manual for new employees

Which statement best describes a challenge in specifying software quality attributes?

- A. Quality attributes are universally understood and applied.
- B. All quality attributes are easy to quantify.
- C. **Some quality attributes are difficult to specify due to their abstract nature.**
- D. Quality attributes have no impact on software performance.

What is a sign-off in the context of the review meeting?

- A) A final report summarizing the review
- B) An approval process that requires the manager's signature
- C) **A document indicating participation and concurrence with the review team's findings**
- D) A record of all the errors found during the review

What does a lack of conformance to software requirements indicate?

- A. The software is user-friendly.
- B. The software is incomplete.
- C. **The software lacks quality.**
- D. The software is innovative.

What is the purpose of a Review Issues List?

- A) **To identify and track problem areas within the product**
- B) To provide a list of successful features in the product
- C) To replace the review summary report
- D) To document the overall project timeline

How does Software Quality Assurance (SQA) contribute to customer satisfaction?

- A. By increasing the cost of the software
- B. By reducing software features
- C. By ensuring higher reliability in the software**
- D. By delaying the release of the software

What is one of the primary advantages of implementing Software Quality Assurance (SQA)?

- A. Increased complexity in the development process
- B. Software will have fewer latent defects**
- C. Longer development times
- D. Increased need for manual testing

Which of the following best describes the relationship between faults and failures in software?

- A. A failure is the root cause of all faults.
- B. A fault is a potential cause of a failure, but not all faults result in failures.**
- C. Faults and failures are the same and can be used interchangeably.
- D. A failure always occurs before a fault is identified.

What is one possible outcome if severe errors are found during the review?

- A) The product is accepted without further modification
- B) The product is accepted provisionally with minor corrections
- C) The product is put on hold indefinitely
- D) The product is rejected and another review is scheduled after corrections**

What does a software failure correspond to?

- A. The presence of unused code
- B. A missing feature in the software
- C. A delay in the software release
- D. Erroneous or unexpected runtime behavior observed by a user**

Why is enforcing standards important in software development?

- A. It controls change and ensures consistency during software development and maintenance.**
- B. It eliminates the need for testing.
- C. It increases development costs.
- D. It allows developers to bypass best practices.

What do specified standards define in software development?

- A. A set of testing procedures.
- B. A guide for software design.
- C. Development criteria that guide software engineering.**
- D. The user interface layout.

What is the role of the recorder in the review meeting?

- A) To lead the review meeting

- B) To develop the product being reviewed
- C) To record (in writing) all important issues raised during the review**
- D) To review the product and make notes

Why is Software Quality Assurance (SQA) challenging to implement in small organizations?

- A. Because it requires advanced technology
- B. Because there is a lack of available resources to perform necessary activities**
- C. Because it is only useful in large organizations
- D. Because it requires a large team of experts

What are implicit requirements in software development?

- A. Requirements that are always explicitly documented.
- B. Requirements that pertain only to the software interface.
- C. Requirements that are irrelevant to software quality.
- D. Requirements that are assumed and often go unmentioned.**

What is the consequence if software development criteria specified in standards are not met?

- A. The software will be more secure.
- B. Lack of software quality is likely to result.**
- C. The software will perform faster.
- D. The software will meet user expectations.

How should you pose your written comments during a review?

- A) As direct orders
- B) As questions**
- C) As personal opinions
- D) As suggestions

What must all attendees of the Formal Technical Review (FTR) do at the end of the review meeting?

- A) Provide feedback anonymously
- B) Make a decision regarding the product**
- C) Write a detailed report on the product's quality
- D) Suggest new features for the product

How much time should each person spend on advance preparation for the review meeting?

- A) No more than 30 minutes
- B) No more than 1 hour
- C) No more than 2 hours**
- D) No specific time limit

What must be done with organization-wide policies, procedures, and standards to make them applicable to specific projects?

- A. They must be discarded.

- B. They must be completely rewritten.
- C. They must be tailored to fit the specific project.**
- D. They must be ignored.

What is the first step in ensuring quality in a software product according to the three-prong approach?

- A. Conducting user testing
- B. Hiring external consultants
- C. Developing the software
- D. Establishing organization-wide policies, procedures, and standards**

What does it mean to control quality in a software project?

- A. Ignoring procedures to save time
- B. Ensuring that appropriate procedures are followed for each project**
- C. Letting the development team decide the best practices
- D. Only focusing on testing

What is the purpose of applying technical methods during software development?

- A. To reduce development costs
- B. To help the analyst achieve a high-quality specification and design**
- C. To minimize user involvement
- D. To automate the coding process

What does the tension between software quality attributes like efficiency and reliability suggest?

- A. Software can be both efficient and reliable without compromise.
- B. Improving one attribute may negatively impact another.**
- C. All quality attributes can be maximized simultaneously.
- D. Software development does not require trade-offs.

What is the definition of software reliability?

- A) Probability of failure during installation
- B) Probability of perfect performance in any environment
- C) Probability of failure during development
- D) Probability of failure-free operation for a specified time in a specified environment**

What is the best course of action if the product is accepted without further modification?

- A) Schedule another review to confirm the decision
- B) Move forward with the product's next phase**
- C) Reevaluate the product to ensure no errors were missed
- D) Delay the sign-off until all team members have reviewed the product again

What key information should be included in a Review Summary Report?

- A) The software's release date

- B) The tools used for the review
- C) What was reviewed, who reviewed it, and the findings and conclusions**
- D) The estimated cost of corrections

True or False: Removing X% of faults always results in an X% improvement in reliability.

- A) True
- B) False**

Which of the following standards might be used to ensure software quality?

- A. ISO 9000-3**
- B. TCP/IP standards
- C. HTML5 standards
- D. CSS standards

When are quality reviews typically applied?

- A. Only at the beginning of a project
- B. During and/or at the end of each life cycle phase**
- C. Only after the product is released
- D. Only during the design phase

What should you do after skimming the product material?

- A) Read the product material thoroughly and annotate a hardcopy**
- B) Skip to the review meeting
- C) Focus solely on style issues
- D) Discuss the material with other reviewers before making notes

What is the fundamental purpose of quality reviews?

- A. To validate the quality of a product or process**
- B. To increase project costs
- C. To replace the need for testing
- D. To reduce project duration

What impact does SQA have on maintenance costs?

- A. SQA increases maintenance costs.
- B. SQA has no impact on maintenance costs.
- C. SQA can reduce maintenance costs.**
- D. SQA only affects development costs.

Why do standards exist in the context of software quality assurance?

- A. To limit innovation in software development
- B. To make software development more expensive
- C. To confuse the development team

D. To help an organization draft an appropriate software quality assurance plan

What does the simple definition of software quality imply?

- A. Software functions without bugs.
- B. Software meets its specifications.**
- C. Software is user-friendly.
- D. Software is efficient.

What should you do if you cannot prepare for a review?

- A) Annotate the hardcopy of the material
- B) Inform the review leader**
- C) Skip the review entirely
- D) Focus on issues of style

What happens if software conforms to its explicit requirements but fails to meet its implicit requirements?

- A. Software quality is considered excellent.
- B. Software meets all user expectations.
- C. Software quality is considered questionable.**
- D. Software becomes more adaptable.

What is the effect of SQA on the overall life cycle cost of software?

- A. It increases the overall life cycle cost.
- B. It has no effect on the life cycle cost.
- C. It only affects the initial development cost.
- D. It reduces the overall life cycle cost of software.**

Which statement is correct about software reliability improvements?

- A) Removing any faults will result in a proportional improvement in reliability.
- B) Software reliability improves only when all faults are removed.
- C) Removing faults in the most critical areas is more important than the quantity of faults removed.**
- D) A 60% fault removal always results in a 60% reliability improvement.

What is considered the foundation for measuring software quality?

- A. Development tools
- B. Software requirements**
- C. User feedback
- D. Code reviews

What role can external entities play in ensuring software quality?

- A. They can be contracted to verify that an organization is standard-compliant.**
- B. They can replace the internal quality assurance team.
- C. They can write the software code.

D. They can disregard existing standards.

What does it mean to accept the product provisionally during the review?

- A) The product is accepted, but only with significant rework
- B) The product is accepted, but minor corrections are needed without requiring further reviews**
- C) The product is accepted without any corrections
- D) The product is rejected due to major errors

What is the recommended duration for a review meeting?

- A) Less than one hour
- B) Less than two hours**
- C) Between 2 and 3 hours
- D) As long as needed to cover all topics

Does the presence of a fault in the software always lead to a failure?

- A. Yes, faults always cause failures.
- B. Yes, but only in large software systems.
- C. No, the presence of a fault doesn't necessarily imply the occurrence of a failure.**
- D. No, faults only cause failures in certain programming languages

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Which of the following is a MAJOR task of test implementation and execution?

- A. Measuring and analyzing results.
- B. Reporting discrepancies as incidents.**
- C. Identifying test conditions or test requirements.
- D. Assessing if more tests are needed.

White Box Techniques are also called as:

- A. Structural Testing**
- B. Design Based Testin
- C. Error Guessing Technique
- D. Experience Based Technique

As part of which test process do you determine the exit criteria?

- A. Test planning.**
- B. Evaluating exit criteria and reporting.
- C. Test closure.
- D. Test control.

What determines the level of risk?

- A. The cost of dealing with an adverse event if it occurs.
- B. The probability that an adverse event will occur.
- C. The amount of testing planned before release of a system.

D. The likelihood of an adverse event and the impact of the event.

Which of the following would you NOT usually find on a software incident report?

- A. The name and/or organisational position of the person raising the problem.
- B. Version of the Software Under Test.
- C. Suggestions as to how to fix the problem.**
- D. Actual and expected results

Which of the following is MOST characteristic of specification based (black-box) techniques?

- A. Test cases can be easily automated.
- B. Test cases are independent of each other.
- C. Test cases are derived systematically from models of the system .**
- D. Test cases are derived systematically from the delivered code.

When a defect is detected and fixed then the software should be retested to confirm that the original defect has been successfully removed. This is called_____

- A. Regression testing
- B. Maintenance testing
- C. Confirmation testing**
- D. None of the above

Bug life cycle

- A. Open, Assigned, Fixed, Closed**
- B. Open, Fixed, Assigned, Closed
- C. Assigned, Open, Closed, Fixed
- D. Assigned, Open, Fixed, Closed

Which of these are objectives for software testing?

- A. Determine the productivity of programmers
- B. Eliminate the need for future program maintenance
- C. Eliminate every error prior to release
- D. Uncover software errors**

Contract and regulation testing is a part of

- A. System testing
- B. Acceptance testing**
- C. Integration testing
- D. Smoke testing

‘X’ has given a data on a person age, which should be between 1 to 99. Using BVA which is the appropriate one

- A. 0,1,2,99
- B. 1, 99, 100, 98
- C. 0, 1, 99, 100**

D. -1, 0, 1, 99

Beta testing is performed at developing organization's site where as Alpha testing is performed by people at their own locations.

A. True

B. False

The _____ Testing will be performed by the people at client own locations

A. Alpha testing

B. Field testing

C. Performance testing

D. System testing

Purpose of test design technique is

A. Identifying test conditions only, not Identifying test cases

B. Not Identifying test conditions, Identifying test cases only

C. Identifying test conditions and Identifying test cases

D. Identifying test conditions or Identifying test cases

Boundary value analysis can only be used during white-box testing.

A. True

B. False

Which type of test design techniques does the following statement best describe a procedure to derive test cases based on the specification of a component?

A. Black Box Techniques.

B. White Box Techniques.

C. Glass Box Techniques.

D. Experience Based Techniques.

Which of the following activities should be performed during the selection and implementation of a testing tool?

i) Investigate the organisation's test process.

ii) Conduct a proof of concept.

iii) Implement the selected tool on a project behind schedule to save time.

iv) Identify coaching and mentoring requirements for the use of the selected tool.

A. i, ii, iii.

B. ii, iii, iv.

C. i, iii, iv.

D. i, ii, iv.

A Test Plan Outline contains which of the following:

i. Test Items

ii. Test Scripts

iii. Test Deliverables

iv. Responsibilities

A. i,ii,iii are true and iv is false

B. i,iii,iv are true and ii is false

C. ii,iii are true and i and iv are false

D. i,ii are false and iii , iv are true

Which is the non-functional testing

A. Performance testing

B. Unit testing

C. Regression testing

D. Sanity testing

What is the MAIN benefit of designing tests early in the life cycle?

A. It is cheaper than designing tests during the test phases.

B. It helps prevent defects from being introduced into the code.

C. Tests designed early are more effective than tests designed later.

D. It saves time during the testing phases when testers are busy.

Testing where in we subject the target of the test , to varying workloads to measure and evaluate the performance behaviors and ability of the target and of the test to continue to function properly under these different workloads.

A. Load Testing

B. Integration Testing

C. System Testing

D. Usability Testing

Where may functional testing be performed?

A. At system and acceptance testing levels only.

B. At all test levels.

C. At all levels above integration testing.

D. At the acceptance testing level only.

Which activities form part of test planning?

i) Developing test cases.

ii) Defining the overall approach to testing.

iii) Assigning resources.

iv) Building the test environment

v) Writing test conditions.

A. i, ii & iv are true, iii & v are false.

B. ii & iii are true, i, iv & v are false.

- C. iv & v are true, i, ii & iii are false.
- D. i, ii & iii are true iv & v are false.

System testing should investigate

- A. Non-functional requirements only not Functional requirements
- B. Functional requirements only not non-functional requirements
- C. Non-functional requirements and Functional requirements**
- D. Non-functional requirements or Functional requirements

Which is not a testing principle

- A. Early testing
- B. Defect clustering
- C. Pesticide paradox
- D. Exhaustive testing**

Who would USUALLY perform debugging activities?

- A. Developers.**
- B. Analysts.
- C. Testers.
- D. Incident Managers.

Exhaustive testing is possible

- A. True
- B. False**

Which of the following is a MAJOR task of test planning?

- A. Scheduling test analysis and design tasks.**
- B. Initiating corrective actions.
- C. Monitoring progress and test coverage.
- D. Measuring and analyzing results.

Which of the following is the task of a Tester?

- i. Interaction with the Test Tool Vendor to identify best ways to leverage test tool on the project.
 - ii. Prepare and acquire Test Data
 - iii. Implement Tests on all test levels, execute and log the tests.
 - iv. Create the Test Specifications
- A. i, ii, iii is true and iv is false
 - B. ii,iii,iv is true and i is false**
 - C. i is true and ii,iii,iv are false
 - D. iii and iv is correct and i and ii are incorrect

A _____ is the step-by-step method followed to ensure that standards are met

- A. SDLC**

- B. Project Plan
- C. Policy
- D. Procedure**

A wholesaler sells printer cartridges. The minimum order quantity is 5. There is a 20% discount for orders of 100 or more printer cartridges. You have been asked to prepare test cases using various values for the number of printer cartridges ordered.

Which of the following groups contain three test inputs that would be generated using Boundary Value Analysis?

- A. 5, 6, 20
- B. 4, 5, 80
- C. 4, 5, 99**
- D. 1, 20, 100

What is the normal order of activities in which software testing is organized?

- A. Unit, integration, system, validation**
- B. System, integration, unit, validation
- C. Unit, integration, validation, system
- D. None of the above

Which of the following are characteristic of regression testing ?

- i) Regression testing is run ONLY once
- ii) Regression testing is used after fixes have been made
- iii) Regression testing is often automated
- iv) Regression tests need not be maintained

- A. ii, iv.
- B. ii, iii.**
- C. i, iii, iv.
- D. iii.

Quality Assurance is the process by which product quality is compared with the application standards and the action taken when nonconformance is detected.

- A. Test Harness
- B. Test Suite**
- C. Test Cycle
- D. Test Driver

Which of the following is a purpose of the review planning phase?

- A. Log defects.
- B. Explain the documents to the participants.
- C. Gather metrics.
- D. Allocate the individual roles.**

What is beta testing?

- A. Testing performed by potential customers at the developers location.
- B. Testing performed by potential customers at their own locations.**
- C. Testing performed by product developers at the customer's location.
- D. Testing performed by product developers at their own locations.

Which of the following combinations correctly describes a valid approach to component testing:

- i) Functional testing of the component in isolation.
- ii) Structure-based testing of the code without recording incidents.
- iii) Automated tests that are run until the component passes.
- iv) Functional testing of the interfaces between modules.

- A. i and ii.
- B. i, ii and iii**
- C. iii.
- D. ii and iv

Which of the following is TRUE?

- A. Confirmation testing is testing fixes to a set of defects and Regression testing is testing to establish whether any defects have been introduced as a result of changes.**
- B. Confirmation testing is testing to establish whether any defects have been introduced as a result of changes and Regression testing is testing fixes to a set of defects.
- C. Confirmation testing and Regression testing are both testing to establish whether any defects have been introduced as a result of changes.
- D. Confirmation testing and Regression testing are both testing fixes to a set of defects.

Which of the following defines the expected results of a test?

- A. Test case specification.**
- B. Test design specification.
- C. Test procedure specification.
- D. Test results.

Which is not the testing objectives

- A. Finding defects
- B. Gaining confidence about the level of quality and providing information
- C. Preventing defects.
- D. Debugging defects**

Which of the following is the best source of Expected Outcomes for User Acceptance Test scripts?

- A. Actual results
- B. Program specification
- C. User requirements**
- D. System specification

Alpha testing is:

- A. Post-release testing by end user representatives at the developer's site.
- B. The first testing that is performed.
- C. Pre-release testing by end user representatives at the developer's site.**
- D. Pre-release testing by end user representatives at their sites.

What is the purpose of test exit criteria in the test plan?

- A. To specify when to stop the testing activity**
- B. To set the criteria used in generating test inputs
- C. To ensure that the test case specification is complete
- D. To know when a specific test has finished its execution

User Acceptance Testing

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- A. Same as Alpha Testing
- B. Same as Beta Testing
- C. Combination of Alpha and Beta Testing
- D. None of the above (không biết có sai đề không)**

Enough testing has been performed when:

- A. Time runs out.
- B. The required level of confidence has been achieved.**
- C. No more faults are found.
- D. The users won't find any serious faults.

Integration testing in the small:

- A. Tests the individual components that have been developed.
- B. Tests interactions between modules or subsystems.**
- C. Only uses components that form part of the live system.
- D. Tests interfaces to other systems

Testing Process comprised of

- A. Test Plan and Test Cases
- B. Test log and Test Status
- C. Defect Tracking
- D. All of the above**

A program validates a numeric field as follows:

Values less than 10 are rejected, values between 10 and 21 are accepted, values greater than or equal to 22 are rejected.

Which of the following input values cover all of the equivalence partitions?

- A. 10,11,21
- B. 3,20,21
- C. 3,10,22**
- D. 10,21,22

Objective of review meeting is

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- A. To identify problems with design
- B. To solve the problems with design
- C. Both A. and B**
- D. None of the above.

Regression testing mainly helps in

- A. Re-testing fixed defects
- B. Checking for side-effects of fixes**
- C. Checking the core gaps
- D. Ensuring high level sanity

Test Plan

- A. Road map for testing**
- B. Tells about the actual results and expected results
- C. Both a and b
- D. None of the above

Which of the following is not described in a unit test standard?

- A. Syntax testing
- B. Equivalence partitioning
- C. Stress testing**
- D. Modified condition/decision coverage

Functional tests can be performed at all test levels

- A. True**
- B. False

In a system designed to work out the tax to be paid:

An employee has £4000 of salary tax free.

The next £1500 is taxed at 10%.

The next £28000 after that is taxed at 22%.

Any further amount is taxed at 40%.

To the nearest whole pound, which of these is a valid Boundary Value Analysis test case?

- A. £28000.
- B. £33501.**

- C. £32001.
- D. £1500.

In a flight reservation system, the number of available seats in each plane model is an input. A plane may have any positive number of available seats, up to the given capacity of the plane.

Using Boundary Value analysis, a list of available – seat values were generated. Which of the following lists is correct?

- A. 1, 2, capacity -1, capacity, capacity plus 1
- B. 0, 1, capacity, capacity plus 1**
- C. 0, 1, 2, capacity plus 1, a very large number
- D. 0, 1, 10, 100, capacity, capacity plus one

Which of the following defines the sequence in which tests should be executed?

- A. Test plan.
- B. Test procedure specification.**
- C. Test case specification.
- D. Test design specification

Given the following specification, which of the following values for age are in the SAME equivalence partition?

If you are less than 18, you are too young to be insured. Between 18 and 30 inclusive, you will receive a 20% discount. Anyone over 30 is not eligible for a discount.

- A. 17, 18, 19.
- B. 29, 30, 31.
- C. 18, 29, 30.**
- D. 17, 29, 31.

Black-box testing technique is also called as structure based technique

- A. True
- B. False**

In a system designed to work out the tax to be paid:

An employee has £4000 of salary tax free.

The next £1500 is taxed at 10%.

The next £28000 is taxed at 22%

Any further amount is taxed at 40%

Which of these groups of numbers would fall into the same equivalence class?

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- A. £4800; £14000; £28000
- B. £5200; £5500; £28000**
- C. £28001; £32000; £35000 x
- D. £5800; £28000; £32000

Which of the following would NOT normally form part of a test plan?

- A. Features to be tested
- B. Incident reports**
- C. Risks
- D. Schedule

Preparing and automating test cases before coding is called

- A. Test first approach
- B. Test-driven development**
- C. Both A. & B.
- D. None of the above

Which of the following is a valid collection of equivalence classes for the following problem:
An integer field shall contain values from and including 1 to and including 15

- A. Less than 1, 1 through 15, more than 15**
- B. Negative numbers, 1 through 15, above 15
- C. Less than 1, 1 through 14, more than 15
- D. Less than 0, 1 through 14, 15 and more

The most important thing about early test design is that it:

- A. Makes test preparation easier.
- B. Means inspections are not required.
- C. Can prevent fault multiplication.**
- D. Will find all faults.

Structural Testing

- A. Same as black box testing
- B. Same as white box testing**
- C. Same as functional testing
- D. None of the above.

In prioritising what to test, the most important objective is to:

- A. Find as many faults as possible.
- B. Test high risk areas.**
- C. Obtain good test coverage.
- D. Test whatever is easiest to test.

We split testing into distinct stages primarily because:

- A. Each test stage has a different purpose.**
- B. It is easier to manage testing in stages.
- C. We can run different tests in different environments.
- D. The more stages we have, the better the testing.

Which one of the following statements about system testing is NOT true?

- A. System tests are often performed by independent teams.
- B. Functional testing is used more than structural testing.
- C. Faults found during system tests can be very expensive to fix.
- D. End-users should be involved in system tests.**

Equivalence partitioning consists of various activities:

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- A. Ensure that test cases test each input and output equivalence class at least once
- B. Identify all inputs and all outputs
- C. Identify equivalence classes for each input
- D. All of the above**

Which of the following should NOT normally be an objective for a test?

- A. To find faults in the software.
- B. To assess whether the software is ready for release.
- C. To demonstrate that the software doesn't work. **x**
- D. To prove that the software is correct.**

Which of the following is a form of functional testing?

- A. Boundary value analysis**
- B. Usability testing
- C. Performance testing
- D. Security testing

Evaluating testability of the requirements and system are a part of which phase

- A. Test Analysis and Design**
 - B. Test Planning and control
 - C. Test Implementation and execution
 - D. Evaluating exit criteria and reporting
-

IEEE 829 test plan documentation standard contains all of the following except:

- A. Test items
- B. Test deliverables
- C. Test tasks

- D. Test environment
- E. Test specification**

Minimum Tests Required for Statement Coverage and Branch Coverage:

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```
Read P
Read Q
If p+q > 100 then
Print "Large"
End if
If p > 50 then
Print "pLarge"
End if
```

- A. Statement coverage is 2, Branch Coverage is 2
- B. Statement coverage is 3 and branch coverage is 2 (x)**
- C. Statement coverage is 1 and branch coverage is 2
- D. Statement Coverage is 4 and Branch coverage is 2

Features of White Box Testing Technique:

- i. We use explicit knowledge of the internal workings of the item being tested to select the test data.
- ii. Uses specific knowledge of programming code to examine outputs and assumes that the tester knows the path of logic in a unit or a program.
- iii. Checking for the performance of the application
- iv. Also checks for functionality

- A. i, ii are true and iii and iv are false**
- B. iii is true and i,ii, iv are false
- C. ii ,iii is true and i,iv is false
- D. iii and iv are true and i,ii are false

Expected results are:

- A. Only important in system testing
- B. Only used in component testing
- C. Never specified in advance
- D. Most useful when specified in advance**

E. Derived from the code

Which of the following is NOT part of a high level test plan?

0/1

- A. Functions not to be tested.
- B. Environmental requirements.
- C. Analysis of Specifications.**
- D. Entry and Exit criteria.

Given the following code, which is true:

```
IF A > B THEN  
C = A - B  
ELSE  
C = A + B  
ENDIF  
Read D
```

```
IF C = D Then  
Print "Error"  
ENDIF
```

- A. 1 test for statement coverage, 3 for branch coverage
- B. 2 tests for statement coverage, 2 for branch coverage
- C. 2 tests for statement coverage. 3 for branch coverage**
- D. 3 tests for statement coverage, 3 for branch coverage
- E. 3 tests for statement coverage, 2 for branch coverage x

A typical commercial test execution tool would be able to perform all of the following EXCEPT:

- A. Generating expected outputs**
- B. Replaying inputs according to a programmed script
- C. Comparison of expected outcomes with actual outcomes
- D. Recording test inputs
- E. Reading test values from a data file

Which of the following is not a part of the Test Implementation and Execution Phase

- A. Creating test suites from the test cases
- B. Executing test cases either manually or by using test execution tools
- C. Comparing actual results
- D. Designing the Tests**

Which of the following is not included in Test Plan

0/1

- A. Features to be tested.
- B. Environmental needs.
- C. Suspension criteria.**
- D. Expected results.

Minimum Test Required for Statement Coverage:

```
Disc = 0
Order-qty = 0
Read Order-qty
If Order-qty >=20 then
  Disc = 0.05
If Order-qty >=100 then
  Disc =0.1
End if
End if
```

- A. Statement coverage is 4
- B. Statement coverage is 1**
- C. Statement coverage is 3
- D. Statement Coverage is 2

Software quality is not relevant to _____

- A. Correctness
- B. Usability
- C. Viability**
- D. Reusability

Match the following:

1. Test estimation
2. Test control
3. Test monitoring

a. Measures of tracking process

- b. Effort required to perform activities
- c. Reallocation of resources

A. 1-b, 2-c, 3-a

- B. 1-b, 2-a, 3-c
- C. 1-c, 2-a, 3-b
- D. 1-a, 2-b, 3-c

Which of the following has highest level of independence in which test cases are:

- A. Designed by persons who write the software under test
- B. Designed by a person from a different section
- C. Designed by a person from a different organization**
- D. Designed by another person

Given the following code, which is true about the minimum number of test cases required for full statement and branch coverage:

```
Read P
Read Q
IF P+Q > 100 THEN
Print "Large"
ENDIF
```

```
If P > 50 THEN
Print "P Large"
ENDIF
```

- A. 1 test for statement coverage, 3 for branch coverage
- B. 1 test for statement coverage, 2 for branch coverage**
- C. 1 test for statement coverage, 1 for branch coverage
- D. 2 tests for statement coverage, 3 for branch coverage
- E. 2 tests for statement coverage, 2 for branch coverage

How many test cases are necessary to cover all the possible sequences of statements (paths) for the following program fragment? Assume that the two conditions are independent of each other:

```
if (Condition 1)
then statement 1
else statement 2
fi
if (Condition 2)
then statement 3
fi
```

0/1

- A. 2 Test Cases
- B. 3 Test Cases
- C. 4 Test Cases**
- D. Not achievable

Which of the following is not a major task of Exit criteria?

- A. Checking test logs against the exit criteria specified in test planning.
- B. Logging the outcome of test execution.**
- C. Assessing if more tests are needed.
- D. Writing a test summary report for stakeholders.

During which test activity could faults be found most cost effectively?

0/1

- A. Execution
- B. Design**
- C. Planning
- D. Check Exit criteria completion

Test planning has which of the following major tasks?

- i. Determining the scope and risks, and identifying the objectives of testing.
 - ii. Determining the test approach (techniques, test items, coverage, identifying and interfacing the teams involved in testing , testware)
 - iii. Reviewing the Test Basis (such as requirements, architecture, design, interface)
 - iv. Determining the exit criteria.
- A. i,ii,iv are true and iii is false**
 - B. i,,iv are true and ii is false
 - C. i,ii are true and iii,iv are false
 - D. ii,iii,iv are true and i is false

Regression testing should be performed:

- i. Every week
- ii. After the software has changed

- iii. As often as possible
- iv. When the environment has changed
- v. When the project manager says

- A. i & ii are true, iii, iv & v are false
- B. ii, iii & iv are true, i & v are false
- C. ii & iv are true, i, iii & v are false**
- D. ii is true, i, iii, iv & v are false

Repeated Testing of an already tested program, after modification, to discover any defects introduced or uncovered as a result of the changes in the software being tested or in another related or unrelated software component:

- A. Re Testing
- B. Confirmation Testing
- C. Regression Testing**
- D. Negative Testing

Which of the following is NOT included in the Test Plan document of the Test Documentation Standard:

- 0/1
- A. Test items (i.e. software versions)
 - B. What is not to be tested
 - C. Test environments
 - D. Quality plans
 - E. Schedules and deadlines**

Which one of the following are non-functional testing methods?

- A. System testing
- B. Usability testing
- C. Performance testing
- D. Both B & C**

Deciding How much testing is enough should take into account :

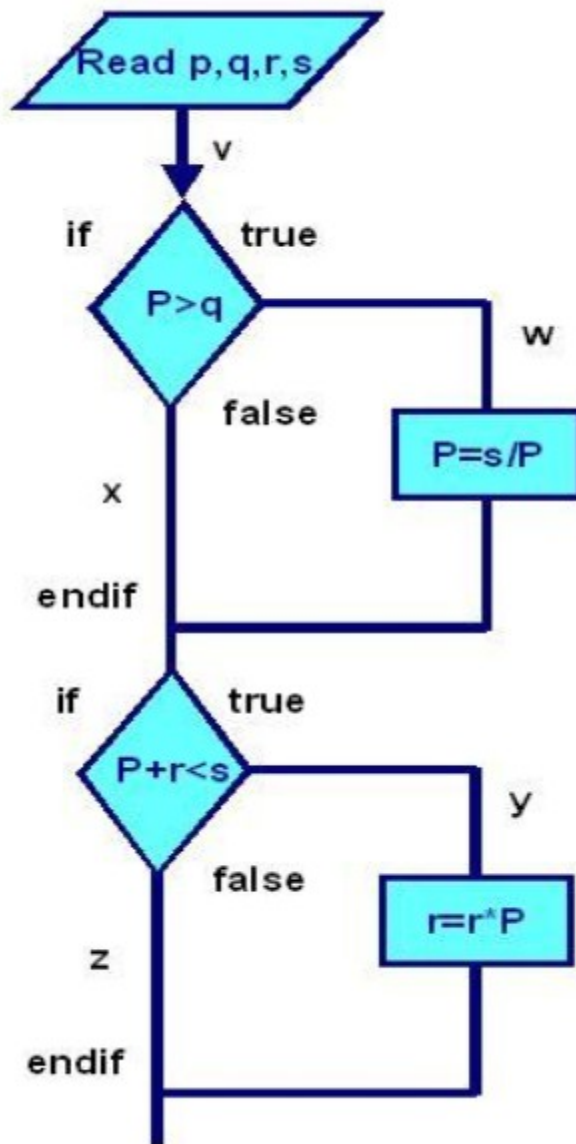
- i. Level of Risk including Technical and Business product and project risk
 - ii. Project constraints such as time and budget
 - iii. Size of Testing Team
 - iv. Size of the Development Team
- A. i,ii,iii are true and iv is false
 - B. i,,iv are true and ii is false
 - C. i,ii are true and iii,iv are false**
 - D. ii,iii,iv are true and i is false

The below diagram represents the following paths through the code

- A. vwy
- B. vwz
- C. vxy
- D. vxz

What is the MINIMUM combination of paths required to provide full statement coverage?

0/1



- A. A
- B. ABD
- C. ABCD
- D. ACD

Given the following:

Switch PC on
Start "outlook"
IF outlook appears THEN
Send an email
Close outlook

- A. 1 test for statement coverage, 1 for branch coverage
- B. 1 test for statement coverage, 2 for branch coverage**
- C. 1 test for statement coverage, 3 for branch coverage
- D. 2 tests for statement coverage, 2 for branch coverage
- E. 2 tests for statement coverage, 3 for branch coverage

The difference between re-testing and regression testing is

- A. Re-testing is running a test again; regression testing looks for unexpected side effects**
- B. Re-testing looks for unexpected side effects; regression testing is repeating those tests
- C. Re-testing is done after faults are fixed; regression testing is done earlier
- D. Re-testing uses different environments, regression testing uses the same environment
- E. Re-testing is done by developers, regression testing is done by independent testers

Beta testing is:

- A. Performed by customers at their own site**
- B. Performed by customers at their software developer's site
- C. Performed by an independent test team
- D. Useful to test bespoke software
- E. Performed as early as possible in the lifecycle

Which of the following is the task of a Test Lead / Leader.

- i. Interaction with the Test Tool Vendor to identify best ways to leverage test tool on the project.
- ii. Write Test Summary Reports based on the information gathered during testing
- iii. Decide what should be automated , to what degree and how.
- iv. Create the Test Specifications

- A. i, ii, iii is true and iv is false**
- B. ii,iii,iv is true and i is false
- C. i is true and ii,iii,iv are false
- D. iii and iv is correct and i and ii are incorrect

One Key reason why developers have difficulty testing their own work is :

- A. Lack of technical documentation
- B. Lack of test tools on the market for developers
- C. Lack of training
- D. Lack of Objectivity**

Regression testing always involves

- A. Testing whether a known software fault been fixed.
- B. Executing a large number of different tests.
- C. Testing whether modifications have introduced adverse side effects.**
- D. Using a test automation tool.

Test Conditions are derived from:

- A. Specifications**
- B. Test Cases
- C. Test Data
- D. Test Design

Which of the following is NOT part of system testing:

- A. Business process-based testing
- B. Performance, load and stress testing
- C. Requirements-based testing
- D. Usability testing
- E. Top-down integration testing**

We use the output of the requirement analysis, the requirement specification as the input for writing:

- A. User Acceptance Test Cases**
- B. Integration Level Test Cases
- C. Unit Level Test Cases
- D. Program specifications

Complete statement and branch coverage means:

- A. That you have tested every statement in the program.
- B. That you have tested every statement and every branch in the program.**
- C. That you have tested every IF statement in the program.
- D. That you have tested every combination of values of IF statements in the program

What is the purpose of test exit criteria in the test plan?

- A. To specify when to stop the testing activity**
- B. To set the criteria used in generating test inputs
- C. To ensure that the test case specification is complete
- D. To know when a specific test has finished its execution

Which of the following project inputs influence testing?

- (I) Contractual requirements**
- (II) Legal requirements(III) Industry standards**

(IV) Application risk

(V) Project size

- A. (I) through (III) are correct
- B. All alternatives are correct**
- C. (II) and (V) are correct
- D. (I), (III) and (V) are correct

Maintenance testing is:

- A. Testing management
- B. Synonym of testing the quality of service
- C. Triggered by modifications, migration or retirement of existing software**
- D. Testing the level of maintenance by the vendor

Why is incremental integration preferred over "big bang" integration?

- A. Because incremental integration has better early defects screening and isolation ability**
- B. Because "big bang" integration is suitable only for real time applications
- C. Incremental integration is preferred over "Big Bang Integration" only for "bottom up" development model
- D. Because incremental integration can compensate for weak and inadequate component testing

V-Model is:

- A. A software development model that illustrates how testing activities integrate with software development phases**
- B. A software life-cycle model that is not relevant for testing
- C. The official software development and testing life-cycle model of ISTQB
- D. A testing life cycle model including unit, integration, system and acceptance phases

Which of the following items need not to be given in an incident report?

- A. The version number of the test object
- B. Test data and used environment
- C. Identification of the test case that failed
- D. The location and instructions on how to correct the fault**

Acceptance testing means

- A. Testing performed on a single stand – alone module or unit of code
- B. Testing after changes have been made to ensure that no unwanted changes were introduced
- C. Testing to ensure that the system meets the needs of the organization and end user.**
- D. Users test the application in the developers environment

The _____ Is the activity where general testing objectives are transformed into tangible test conditions and test designs

- A. Testing Planning

- B. Test Control
- C. Test analysis and design**
- D. Test implementation

A thermometer measures temperature in whole degrees only. If the temperature falls below 18 degrees, the heating is switched off. It is switched on again when the temperature reaches 21 degrees. What are the best values in degrees to cover all equivalence partitions?

- A. 15, 19 and 25.**
- B. 17, 18 and 19.
- C. 18, 20 and 22.
- D. 16, 26 and 32.

The Switch is switched off once the temperature falls below 18 and then it is turned on when the temperature is more than 21. Identify the Equivalence values which belong to the same class.

- A. 12,16,22
- B. 24,27,17
- C. 22,23,24**
- D. 14,15,19

What is an equivalence partition (also known as an equivalence class)?

- A. A set of test cases for testing classes of objects
- B. An input or output range of values such that only one value in the range becomes a test case**
- C. An input or output range of values such that each value in the range becomes a test case
- D. An input or output range of values such that every tenth value in the range becomes a test case.

One of the fields on a form contains a text box which accepts alpha numeric values. Identify the Valid Equivalence class

- A. BOOK
- B. Book
- C. Boo01k**
- D. book

In a Examination a candidate has to score minimum of 24 marks in order to clear the exam. The maximum that he can score is 40 marks. Identify the Valid Equivalence values if the student clears the exam.

- A. 22,23,26
- B. 21,39,40
- C. 29,30,31**
- D. 0,15,22

One of the fields on a form contains a text box which accepts numeric values in the range of 18 to 25. Identify the invalid Equivalence class

- A. 17
- B. 19
- C. 24
- D. 21

Equivalence testing divides the input domain into classes of data from which test cases can be derived to reduce the total number of test cases that must be developed.

- A. True
- B. False

Equivalence partitioning is:

- A. A black box testing technique used only by developers
- B. A black box testing technique than can only be used during system testing
- C. A black box testing technique appropriate to all levels of testing
- D. A white box testing technique appropriate for component testing

Order numbers on a stock control system can range between 10000 and 99999 inclusive. Which of the following inputs might be a result of designing tests for only valid equivalence classes and valid boundaries:

- A. 1000, 5000, 99999
- B. 9999, 50000, 100000
- C. 10000, 50000, 99999
- D. 10000, 99999
- E. 9999, 10000, 50000, 99999, 10000

One of the fields on a form contains a text box, which accepts alphabets in lower or upper case. Identify the invalid Equivalence class value.

- A. CLASS
- B. cLASS
- C. CLass
- D. CLa01ss

In a system designed to work out the tax to be paid:

An employee has £4000 of salary tax free. The next £1500 is taxed at 10% The next £28000 is taxed at 22% Any further amount is taxed at 40% Which of these groups of numbers would fall into the same equivalence class?

- A. £4800; £14000; £28000
- B. £5200; £5500; £28000
- C. £28001; £32000; £35000
- D. £5800; £28000; £32000

If a candidate is given an exam of 40 questions, should get 25 marks to pass (61%) and should get 80% for distinction, what is equivalence class.

- A. 23, 24, 25
- B. 0, 12, 25
- C. 30, 36, 39
- D. 32, 37, 40

Which of these groups of numbers would fall into the same equivalence class?

In a system designed to work out the tax to be paid: An employee has \$4000 of salary tax free. The next \$1500 is taxed at 10%. The next \$28000 is taxed at 22%. Any further amount is taxed at 40%

- A. \$5800; \$28000; \$32000
- B. \$0; \$200; \$4200
- C. \$5200; \$5500; \$28000
- D. \$28001; \$32000; \$35000**

Which of the following is a black box design technique?

- A. Statement testing
- B. Equivalence partitioning**
- C. Error- guessing
- D. Usability testing

A program validates a numeric field as follows:

Values less than 10 are rejected, values between 10 and 21 are accepted, values greater than or equal to 22 are rejected. Which of the following input values cover all of the equivalence partitions?

- A. 10, 11, 21
- B. 3, 20, 21
- C. 3, 10, 22**
- D. 10, 21, 22

Which of the following is a valid collection of equivalence classes for the following problem: An integer field shall contain values from and including 1 to and including 15

- A. Less than 1, 1 through 15, more than 15**
- B. Negative numbers, 1 through 15, above 15
- C. Less than 1, 1 through 14, more than 15
- D. Less than 0, 1 through 14, 15 and more

Equivalence partitioning consists of various activities:

0/1

- A. Ensure that test cases test each input and output equivalence class at least once
- B. Identify all inputs and all outputs
- C. Identify equivalence classes for each input
- D. All of the above**

Which of the following is not described in a unit test standard?

- A. Syntax testing
- B. Equivalence partitioning
- C. Stress testing**
- D. Modified condition/decision coverage

In a system designed to work out the tax to be paid:

0/1

An employee has £4000 of salary tax free. The next £1500 is taxed at 10%

The next £28000 is taxed at 22%

Any further amount is taxed at 40%

Which of these groups of numbers would fall into the same equivalence class?

- A. £4800; £14000; £28000
- B. £5200; £5500; £28000
- C. £28001; £32000; £35000**
- D. £5800; £28000; £32000

Given the following specification, which of the following values for age are in the SAME equivalence partition?

If you are less than 18, you are too young to be insured. Between 18 and 30 inclusive, you will receive a 20% discount. Anyone over 30 is not eligible for a discount.

- A. 17, 18, 19.
- B. 29, 30, 31.
- C. 18, 29, 30.**
- D. 17, 29, 31.

What is the main purpose of impact analysis for testers?

- A. To determine the programming effort needed to make the changes.
- B. To determine what proportion of the changes need to be tested.
- C. To determine how much the planned changes will affect users.
- D. To determine how the existing system may be affected by changes.**

Which of the following is a major task of test planning?

- A. Determining the test approach.**
- B. Preparing test specifications.
- C. Evaluating exit criteria and reporting.
- D. Measuring and analyzing results.

=====

The testing in which code is checked

- Blackbox testing
- Whitebox testing**

Redbox testing
Greenbox testing

What is the primary focus of acceptance testing?

Verifying individual units or components

Ensuring the software meets the user's requirements

Testing the system's behavior under extreme conditions

Evaluating the performance of the testing team

What is the role of "Defect Life Cycle" in software testing?

To evaluate the user interface and overall user experience

To document the testing process

To manage defects from discovery to resolution

To optimize code for better performance

How does "Stress Testing" contribute to software quality assurance?

By optimizing code for better performance

By assessing the software's behavior under varying workloads

By focusing solely on manual testing

By ignoring potential risks in the software

How does "User Acceptance Testing (UAT)" contribute to the testing process?

By evaluating the user interface and overall user experience

By testing the software in a real-world environment before release

By focusing solely on manual testing

By automating all testing processes

What does the term "Test Execution" refer to in the testing process?

Designing test cases

Evaluating the user interface and overall user experience

Performing actual testing by executing test cases

Managing project schedules

How does "Acceptance Criteria" contribute to the testing process?

By evaluating the user interface and overall user experience

By specifying conditions that must be met for a feature to be considered complete

By managing project schedules

By focusing solely on manual testing

What does the term "Test Case" refer to in software testing?

A defect in the code

A tool used for testing

A set of conditions, inputs, and expected results

A type of testing technique

What is the role of a "Test Plan" in testing?

To execute test cases manually

To document test results

To outline the overall testing approach and activities

To optimize software performance

What is the purpose of "Regression Testing" in testing?

To test a new feature

To validate that new code changes haven't negatively impacted existing functionalities

To find defects in the code

To execute test cases manually

How does "Non-Functional Testing" differ from "Functional Testing" in manual testing?

Non-Functional Testing focuses on testing the application's functionality, while Functional Testing assesses non-functional aspects

Non-Functional Testing assesses non-functional aspects, such as performance and usability, while Functional Testing focuses on testing the application's functionality

They are synonymous terms

Both focus solely on manual testing

What is static testing?

Testing the application in a static environment

Testing without executing the code

Testing the user interface

Testing the performance of the application

What is boundary testing?

Testing around the boundary of the application

Testing the limits of input values

Testing the user interface

Testing the security boundaries of the system

What is the purpose of a defect life cycle?

To identify defects

To classify and manage defects from discovery to resolution

To execute test cases

To prioritize test cases

What is the purpose of a test summary report?

To document defects

To provide a summary of testing activities and results

To execute test cases

To analyze code coverage

What is Black Box Testing?

Testing the internal logic and code structure

Testing the software without knowing its internal code

Testing the user interface only

Testing the security vulnerabilities

What is the primary goal of Black Box Testing?

To validate the internal code structure

To find security vulnerabilities

To verify the functionality of the software

To review the source code

What is white box testing?

Testing the software's user interface

Testing the software's internal logic and code structure

Testing the software's performance

Testing the software for security vulnerabilities

What is the primary focus of white box testing?

Functionalities of the software

User interface design

Code execution paths and internal structures

System integration testing

Why is white box testing often referred to as "glass box" testing?

Because the code is visible like glass

Because it only tests the user interface

Because it is fragile like glass

What does the term "path testing" mean in white box testing?

Testing the user's journey through the software

Testing the paths mentioned in the project documentation

Ensuring each possible path through the code is executed

Testing the paths users might take in the software

What is branch coverage in white box testing?

Ensuring all branches of the code are executed

Checking for branches in the project documentation

Testing the branching logic in the user interface

Analyzing the number of branches in the project plan

What is the primary goal of software testing?

Writing code

Debugging

Finding defects in the software

Documenting requirements

Which testing level focuses on the smallest unit of a software system?

System testing

Integration testing

Unit testing

Acceptance testing

What is regression testing?

Testing the same functionality repeatedly

Testing modified code to ensure existing functionalities still work

Testing only new features

Ignoring defects in the code

What is the primary goal of "Alpha Testing"?

To find defects in the code

To evaluate the user interface and overall user experience

To assess the system's performance under varying workloads

To test the software in a controlled environment before releasing it to a limited audience

What is the purpose of equivalence partitioning in software testing?

To divide input values into equivalent classes to reduce test cases

To test the equivalence of system components

To assess the equivalence of different software versions

To find defects in the code

What is the primary goal of Software Quality Assurance (SQA)?

Debugging

Testing

Quality improvement

Code optimization

What does the acronym 'ISO' stand for in the context of software quality?

International Software Organization

Information Security Operations

International Organization for Standardization

Internal System Operations

Which testing type is performed to verify if the individual units or modules of the software work as expected?

System testing

Unit testing

Integration testing
Acceptance testing

Which of the following is a key principle of SQA?

Delay testing until the end of the development process

Correcting defects is more costly in the later stages of development

Testing is only the responsibility of the testing team

Quality is solely the concern of the development team

What is the purpose of the IEEE 829 standard?

Software development process

Test documentation

Code optimization techniques

Requirement analysis

What is the significance of a test strategy in software testing?

It defines the scope and approach of testing

It is a tool for executing test cases

It focuses on the design of the software

What is the primary purpose of a test execution tool?

To design test cases

To execute test cases and compare actual results with expected results

To manage project schedules

To analyze code complexity

What is the primary purpose of a test closure report?

To execute test cases

To document the testing process

To provide a summary of completed testing activities and the current status of the project

To evaluate the user interface

What is the concept of "Test Driven Development" (TDD)?

Writing tests after the code is developed

Writing tests before developing the code

Running tests randomly during development

Ignoring testing in the development process

The Software Requirement Specification(SRS) is said to be _____ if and only if no subset of individual requirements described in it conflict with each other.

Correct

Consistent

Unambiguous

Verifiable

What is the primary purpose of Load Testing in software development?

To identify and fix security vulnerabilities

To measure the performance of a system under expected load

To ensure proper user authentication

To test the functionality of individual components

What is the main metric measured during Load Testing?

Response time

Number of test cases executed

Code coverage

Number of user stories completed

What is the purpose of endurance testing in the context of Load Testing?

To measure the system's performance under peak load conditions

To evaluate the system's ability to handle sustained load over an extended period

To identify and fix security vulnerabilities

To assess the functionality of individual components

Why is it important to monitor server metrics during Load Testing?

To identify and fix security vulnerabilities

To assess the functionality of individual components

To measure the server's performance under different loads

To evaluate the response time of the system

What is the purpose of spike testing in Load Testing scenarios?

To evaluate the system's ability to handle sustained load over an extended period

To measure the system's performance under normal load conditions

To assess the system's response to sudden and extreme increases in load

To identify and fix security vulnerabilities

What is the difference between stress testing and load testing?

Stress testing focuses on peak loads, while load testing simulates expected loads.

Load testing aims to find software defects, while stress testing assesses hardware limitations.

There is no difference; the terms are used interchangeably.

Stress testing is only applicable to web applications, while load testing is for any software.

What is Gray Box Testing?

Testing the software without any knowledge of its internals

Combining elements of both Black Box and White Box Testing

Testing only the user interface

Testing for security vulnerabilities only

Why is Gray Box Testing considered a bridge between Black Box and White Box Testing?

Because it only focuses on user interface testing

Because it provides complete knowledge of the internal code

Because it combines aspects of both without full disclosure

Because it excludes functional testing